

375 km, Unrepeated Transmission Using a Erbium-Doped, Tellurite Fiber Based, 69 nm bandwidth, ROPA

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Abstract We use a 69 nm bandwidth ROPA based on an erbium-doped tellurite glass fiber, combined with bidirectional distributed Raman amplification, to transmit 333 x 24.5 Gbaud channels over 375 km with a minimum span loss of 56 dB, achieving a total throughput of 40.9 Tb/s. ©2026 The Author(s)

Introduction

Unrepeated links use no electrically powered mid-span components and can extend over several hundred kilometers, having demonstrated transmission capacities in the tens of Tb/s range as can be seen in figure 1 [1–18]. Unrepeated transmission systems typically rely on distributed Raman amplification (DRA) to extend the transmission distance, and can be broadly categorized into systems with and without the use of remote optically pumped amplifiers (ROPA). The latter can achieve wide bandwidths of up to 121 nm over S-, C- and L-band, supported by broadband DRA [9]. In contrast, ROPA-assisted systems typically rely on silica-based erbium-doped fibers (EDF), limiting the usable gain bandwidth to 45 nm or less [13–16]. Alternative host materials such as fluoride or tellurite glasses offer significantly broader 3-dB gain bandwidths of up to 90 nm [19–21]. Despite this advantage, such materials have received little attention for unrepeated transmission. Alternatively, the ROPA bandwidth has been extended by the use of separate gain fibers optimized for the C and L band, along with a dedicated pump delivery fiber. This technique has enabled C+L ROPA systems with bandwidths up to 75 nm, at

the cost of increased system complexity [18].

In this work we demonstrate a 69 nm bandwidth erbium-doped tellurite glass fiber (EDTF) based ROPA, which is used to mitigate loss in a 375 km unrepeated transmission link. The ROPA has a wideband gain of 10 dB in the range from 1538 nm to 1607 nm, peaking at a wavelength of 1552 nm with a gain of 23 dB. To further mitigate the link loss, bidirectional DRA is utilized with a pump power of up to 2.5 W. The transmitted signal is composed of 333 x 24.5 Gbaud channels with 25 GHz spacing, based on polarization-multiplexed 4-ary or 16-ary quadrature amplitude modulation (QPSK and 16QAM), and achieves a net data rate above 40 Tb/s. This is the widest bandwidth of a single gain-fiber ROPA reported to date and shows the potential of non-standard glasses in gain fibers used for ROPAs.

Experimental Setup

The experimental setup is shown in figure 2. A test channel was generated by a 65 GSa/s arbitrary waveform generator (AWG) synthesizing a signal at 24.5 Gbaud, which drove a dual-polarization IQ-modulator (IQM) illuminated by an external cavity laser (ECL). Two additional neighboring dummy channels at a 25 GHz spacing were generated by a secondary modulator fed from the same AWG and were subsequently decorrelated. The three channels were then amplified by either a C- or L-band erbium-doped fiber amplifier (EDFA). A broadband amplified spontaneous emission (ASE) source was used to emulate a fully loaded transmission band. The ASE noise generation was followed by optical processors (OP) for the C- and L-band, which flattened the ASE noise and carved a 75 GHz notch to accommodate the test band. The dummy band and test channels were combined via a 10 dB coupler, with per-channel power control implemented by variable optical attenuators (VOAs). The formed dummy band can be seen in figure 3a, spanning from 1538 nm to 1607 nm, with a 2 nm gap between C- and L-band. The band

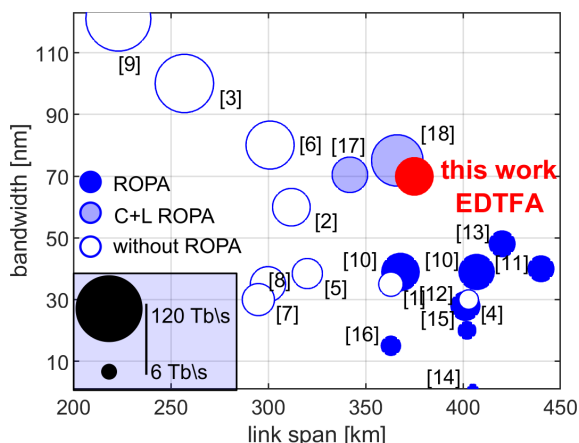


Fig. 1: Transmission bandwidth, distance and data rate of unrepeated links, differentiating between systems using ROPA, a double gain fiber ROPA and systems without ROPA.

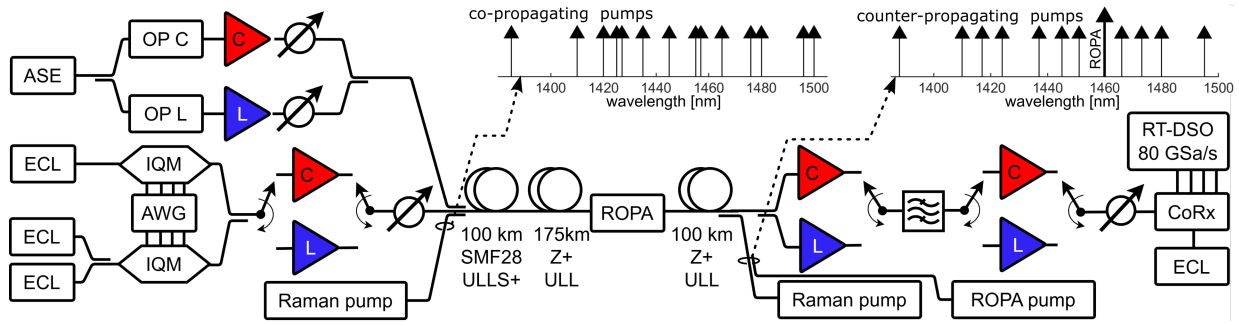


Fig. 2: The setup used for the unrepeated EDTF-ROPA transmission, spanning 375 km.

was launched at a power of 16 dBm and had a linear pre-tilt of approximately 8 dB applied to counteract stimulated Raman scattering (SRS) and to minimize nonlinear impairments on the transmitted channels. The transmission link consisted of a 100 km span of Corning SMF-28 ULLS+, to increase the gain of the co-propagating DRA, followed by 275 km of Sumitomo Z+ULL large effective area fiber ($150 \mu\text{m}^2$), as shown in figure 2, resulting in a total span loss of approximately 56 dB at 1550 nm. The ROPA was placed at a distance of 275 km from the transmitter and consisted of 2 m EDTF. Co-propagating DRA used 11 pumps with wavelengths between 1380 nm and 1500 nm and a combined power of 2.5 W, with pump wavelengths shown in figure 2. Counter-propagating DRA used 11 pump wavelengths from 1410 nm to 1495 nm with a combined pump power of 1.5 W. The ROPA was pumped by a counter-propagating 1460 nm pump at 1 W, supplemented by a 160 mW second-order pump at 1385 nm. Residual counter-propagating DRA pump power additionally contributed to pumping of the ROPA, which benefited from the distributed pump delivery [19]. All pump powers were optimized to maximize gain uniformity across all channels, subject to the upper limit imposed by nonlinear impairments at high pump

powers. The spectral evolution along the link is shown in figure 3a. The co-propagating DRA compensates the pre-applied launch tilt, yielding an approximately flat spectrum at the ROPA input. The ROPA introduces a pronounced spectral non-uniformity: the power decreases by 5 dB from 1550 nm toward 1538 nm at the short-wavelength edge, and by 10 dB from 1552 nm to 1607 nm at the long-wavelength edge, resulting in a negatively tilted spectrum at the ROPA output. The counter-propagating DRA introduces no significant additional tilt, compensating only the residual link loss between the ROPA output and the fiber end, with the exception of the range 1595 nm–1607 nm, where the power decreases sharply by 10 dB. The resulting net loss at the fiber output ranged from approximately 10 dB at 1607 nm to a minimum of 2 dB at 1550 nm, relative to the ROPA output. The corresponding gain profiles are shown in figure 3b. The co-propagating DRA exhibits a minimum gain of 10 dB at 1538 nm, rising to 18 dB at 1585 nm, where it reaches a plateau toward the long-wavelength edge. The ROPA gain reached approximately 23 dB near 1550 nm, limited by the available pump power after propagation through 100 km of fiber, decreasing to approximately 12 dB across the range 1575 nm–1607 nm. The gain peak was observed to shift toward shorter wavelengths with increasing pump power.

At the fiber output, the WDM (wavelength division multiplexed) signal was split into C- and L-band, amplified by low-noise amplifiers, and filtered by a 0.4 nm optical bandpass filter to select the channel under test. The channel was then re-amplified and power-adjusted by a VOA before detection at a coherent receiver (CoRx), using a 60 kHz linewidth laser as the local oscillator. The received signal was digitized by an 8-bit, 33 GHz real-time sampling oscilloscope (DSO) operating at 80 GSa/s. Five 10 μs records were captured per channel, and offline DSP was performed at 2 samples/symbol, comprising clock recovery, chromatic dispersion compensation, frequency offset estimation, matched filtering, and 33-tap 2×2 MIMO equalization for polarization demultiplexing with embedded carrier phase recovery [22]. The MIMO equalizers were initial-

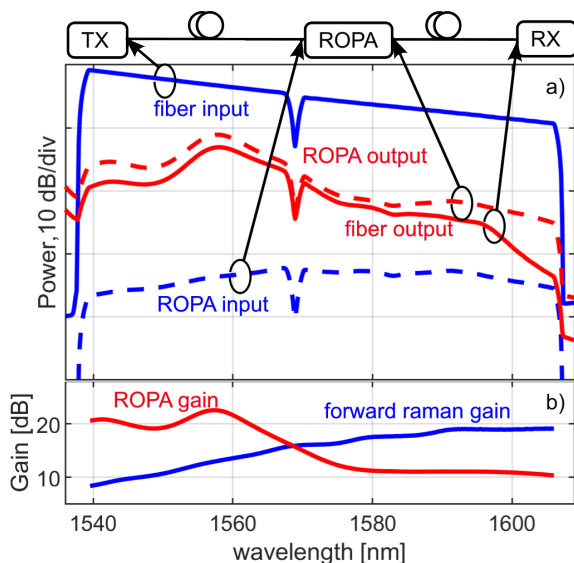


Fig. 3: a) Transmitted signal spectra along different lengths of the transmission link. b) Gain of the co propagating DRA and ROPA.

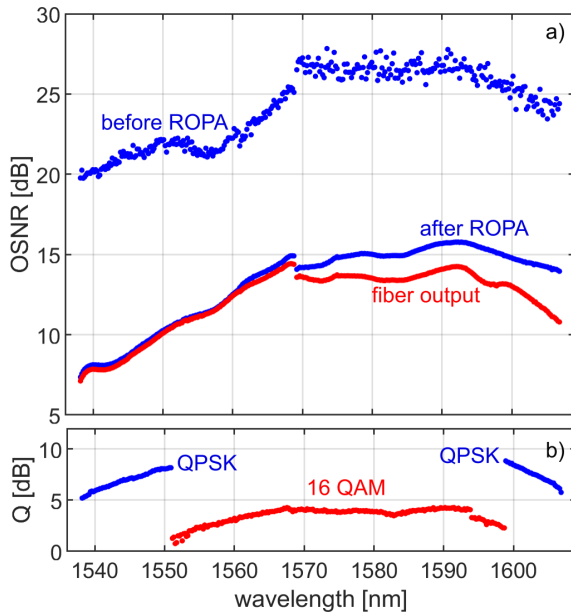


Fig. 4: a) OSNR of the transmitted signal at different points of the transmission link. b) Q-factor of the transmitted signal.

ized in data-aided least-mean squares mode and transitioned to decision-directed operation upon convergence. The OSNR evolution along the link is shown in figure 4a. Prior to the ROPA, after 275 km of propagation, the OSNR ranged from 20 dB–28 dB, with a flat plateau of 28 dB between 1570 nm–1590 nm. Data is subject to considerable noise, due to low signal power at 275 km. Toward longer wavelengths, the OSNR decreases to 23 dB above 1590 nm, while at shorter wavelengths it falls more steeply, reaching 21 dB at 1555 nm. Discontinuities at 1570 nm arise from the use of separate transmitter EDFAs, propagating through all OSNR measurements. Following ROPA amplification, the OSNR peaks at 17 dB near 1590 nm and decreases approximately linearly toward shorter wavelengths, reaching a minimum of 7 dB at 1538 nm. Figure 4a also shows the measured OSNR at the fiber output, where in the C-band region from 1538 nm–1570 nm, the counter-propagating DRA introduced negligible additional OSNR penalty. However, between 1570 nm–1607 nm, the OSNR at the fiber output was degraded relative to the ROPA output, by up to 9 dB, showing close correspondence to the

measured power in the L-band.

Following offline processing, the Q-factor was evaluated for all channels, as shown in figure 4b. At the spectral extremes, where the received OSNR was insufficient for reliable 16QAM recovery, the modulation format was switched to QPSK. The resulting Q-factors ranged from 5 dB–8 dB for QPSK channels and 1.5 dB–4 dB for 16QAM channels, closely tracking the OSNR distribution across the band. Both metrics exhibit a flat plateau in the range 1565 nm–1592 nm, with a monotonic decrease toward the band edges, confirming that the Q-factor performance is predominantly OSNR-limited throughout the transmission band.

Figure 5 shows the estimated data rates of the 333 channels. After signal recovery, the throughput was estimated using the method of [23], based on soft-decision forward error correction (FEC) with punctured DVB-S2 codes to achieve finer code rate granularity. QPSK channels achieved per-channel data rates of 81 Gb/s–97 Gb/s, while 16QAM channels yielded 104 Gb/s–150 Gb/s, closely tracking the Q-factor distribution across the band. The aggregate FEC-decoded throughput reached 40.9 Tb/s, against a GMI-estimated upper bound of 43.75 Tb/s.

These results demonstrate the viability of an EDTF as a single wideband ROPA gain fiber, achieving transmission bandwidths comparable to those of DRA-only systems or dual-fiber C + L band ROPA architectures, while maintaining the simplicity of a single gain fiber.

Conclusions

We have demonstrated an unrepeated transmission link using an EDTF ROPA with a bandwidth of 69 nm. The ROPA is used to support a 375 km link, where we transmit a signal consisting of 333 x 24.5 Gbaud polarization-multiplexed channels, with a total throughput of 40.9 Tb/s. This shows the viability of tellurite-fiber-based ROPA as a solution for wideband unrepeated transmission.

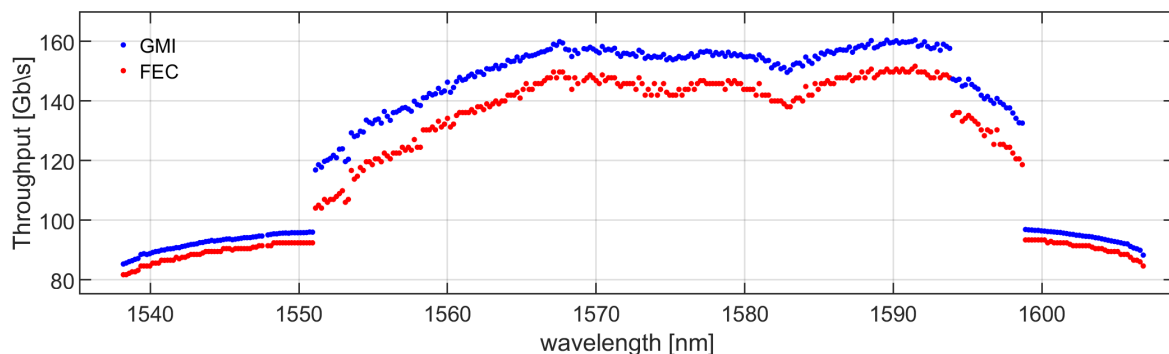


Fig. 5: FEC vs. GMI encoded data rate of the transmitted signal.

Acknowledgements

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References

- [1] D. Orsuti et al., "16.1 tb/s, 363 km, unrepeated c-band transmission with bidirectional raman amplification without ROPA", in *ECOC 2024; 50th European Conference on Optical Communication*, Sep. 2024, pp. 1801–1804.
- [2] R. S. Luis et al., "40.5 tb/s, 312.9 km, unrepeated transmission using erbium-doped tellurite fiber amplifiers", in *49th European Conference on Optical Communications (ECOC 2023)*, vol. 2023, Oct. 2023, pp. 656–659. DOI: 10.1049/icp.2023.2283
- [3] A. Ghazisaeidi et al., "99.35 tb/s ultra-wideband unrepeated transmission over 257 km using semiconductor optical amplifiers and distributed raman amplification", *Journal of Lightwave Technology*, vol. 40, no. 21, pp. 7014–7019, Nov. 2022, ISSN: 1558-2213. DOI: 10.1109/JLT.2022.3198518
- [4] R. S. Luis et al., "403 km, 10.3 tb/s, unrepeated link using incoherent raman amplification without ROPA", in *Optical Fiber Communication Conference (OFC) 2025 (2025)*, paper M1B.2, Optica Publishing Group, Mar. 30, 2025. DOI: 10.1364/OFC.2025.M1B.2
- [5] B. Zhu et al., "25.6 tbit/s (64x400gb/s) real-time unrepeated transmission over 320 km SCUBA fibres by 400zr+ pluggable modules", in *2021 European Conference on Optical Communication (ECOC)*, Sep. 2021, pp. 1–4. DOI: 10.1109/ECOC52684.2021.9606127
- [6] I. F. De Jauregui Ruiz et al., "66.8 tb/s real-time c+l unrepeated transmission over 301 km using forward and backward raman amplification", in *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2024, pp. 1–3. DOI: 10.1364/ofc.2024.w3f.4
- [7] H. Bissessur et al., "Unrepeated transmission of 29.2 tb/s over 295 km with probabilistically shaped 64 QAM", in *2018 European Conference on Optical Communication (ECOC)*, Sep. 2018, pp. 1–3. DOI: 10.1109/ECOC.2018.8535502
- [8] B. Alexis, B. Hans, H. Farana, T. Alexandru, and H. Alain, "Unrepeated c-band transmission of 35.1 tb/s capacity over 300 km using real time 125 GBd PCS-64-QAM", in *ECOC 2024; 50th European Conference on Optical Communication*, Sep. 2024, pp. 1817–1820. DOI: 10.1109/ACP/IPOC63121.2024.10810129
- [9] J. Yang et al., "122.6 tb/s s+c+l band unrepeated transmission over 223 km link with optimized bidirectional raman amplification", *Journal of Lightwave Technology*, vol. 43, no. 4, pp. 1893–1901, Feb. 2025, ISSN: 1558-2213. DOI: 10.1109/JLT.2024.3521233
- [10] A. Busson, H. Bissessur, F. Hedaraly, A. Calsat, and P. Plantady, "Real time c-band unrepeated transmission of 36.4 tb/s in pcs-64qam and 32 tb/s in pcs-16qam over 368 km and 407 km respectively", in *2025 European Conference on Optical Communications (ECOC)*, Sep. 2025, pp. 1–4. DOI: 10.1109/ECOC66593.2025.11263270
- [11] J. Li et al., "80x200 gbit/s real-time un-repeated transmission over 440 km with 2.9-b/s/hz spectral efficiency", in *2020 Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC)*, Oct. 2020, pp. 1–3. DOI: 10.1364/ACP.2020.M4D.1
- [12] Y.-K. Huang et al., "20.7-tb/s repeater-less transmission over 401.1-km using QSM fiber and XPM compensation via transmitter-side DBP", in *2016 21st OptoElectronics and Communications Conference (OECC) held jointly with 2016 International Conference on Photonics in Switching (PS)*, Jul. 2016, pp. 1–3.
- [13] J. Li et al., "16 tbit/s real-time unrepeated transmission over 420 km g.654e fiber", in *2020 European Conference on Optical Communications (ECOC)*, Dec. 2020, pp. 1–3. DOI: 10.1109/ECOC48923.2020.9333042
- [14] H. Bissessur, C. Bastide, S. Dubost, S. Etienne, F. Hedaraly, and M. Duval, "Record single fibre unrepeated transmission of 2 x 500 gb/s over 405.7 km", in *45th European Conference on Optical Communication (ECOC 2019)*, Sep. 2019, pp. 1–4. DOI: 10.1049/cp.2019.0801
- [15] B. Zhu et al., "Unrepeated transmission of 6.3 tb/s (63x128) gb/s over 402-km fiber link", *IEEE Photonics Technology Letters*, vol. 26, no. 17, pp. 1711–1714, Sep. 2014, ISSN: 1941-0174. DOI: 10.1109/LPT.2014.2331338
- [16] H. Bissessur, C. Bastide, S. Dubost, S. Etienne, and D. Mongardien, "8 tb/s unrepeated transmission of real-time processed 200 gb/s PDM 16-QAM over 363 km", in *2014 The European Conference on Optical Communication (ECOC)*, ISSN: 1550-381X, Sep. 2014, pp. 1–3. DOI: 10.1109/ECOC.2014.6963861
- [17] M. Tan et al., "Real-time unrepeated c+l band transmission of 36 tb/s capacity over 342.5 km", in *2023 21st International Conference on Optical Communications and Networks (ICOON)*, ISSN: 2771-3059, Jul. 2023, pp. 1–3. DOI: 10.1109/ICOON59242.2023.10236400
- [18] J. Xu et al., "Real-time unrepeated transmission of 28.13 pb/s-km with 800 gb/s coherent transceivers and novel bi-directional ROPA", *Journal of Lightwave Technology*, vol. 44, no. 3, pp. 1020–1029, Feb. 1, 2026, ISSN: 0733-8724, 1558-2213. DOI: 10.1109/JLT.2025.3624827
- [19] A. Mori, "Tellurite-based fibers and their applications to optical communication networks", *Journal of the Ceramic Society of Japan*, vol. 116, no. 1358, pp. 1040–1051, 2008, ISSN: 1348-6535, 1882-0743. DOI: 10.2109/jcersj2.116.1040
- [20] R. S. Luis et al., "Wideband unrepeated transmission using doped tellurite-fiber amplifiers", *Journal of Lightwave Technology*, vol. 42, no. 13, pp. 4410–4416, Jul. 2024, ISSN: 1558-2213. DOI: 10.1109/JLT.2024.3372472
- [21] L. Ren, Y. Pan, Y. Fu, Y. Cheng, and L. Yuan, "Bismuth-erbium co-doped fiber amplifier for broadband applications", *Optical Engineering*, vol. 63, no. 7, p. 077101, Jul. 2024, ISSN: 0091-3286, 1560-2303. DOI: 10.1117/1.OE.63.7.077101
- [22] T. Pfau, S. Hoffmann, and R. Noe, "Hardware-efficient coherent digital receiver concept with feedforward carrier recovery for m-QAM constellations", *Journal of Lightwave Technology*, vol. 27, no. 8, pp. 989–999, Apr. 2009, ISSN: 1558-2213. DOI: 10.1109/JLT.2008.2010511
- [23] R. S. Luis, B. J. Puttnam, G. Rademacher, Y. Awaji, and H. Furukawa, "Demonstration of a 90 tb/s, 234.8 km, c+l band unrepeated SSMF link with bidirectional raman amplification", *Optics Express*, vol. 30, no. 8, p. 13114, Apr. 11, 2022, ISSN: 1094-4087. DOI: 10.1364/OE.451948